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Experimental and numerical assessments of underlying natural air movement on PV modules temperature

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ABSTRACT

PV panel installation on rooftops is increasing all around the world. The negative effect of the temperature increment on the performance of the PV panels is known. This study aims to evaluate the effect of the gap between the panels and rooftop on the effectiveness of free natural convection to pick up heat from the PV panel. The experimental and numerical investigation was carried out to study the effect of the natural air movement beneath the five PV panels merged vertically. The scope of the study was limited to the temperature measurement of air inlet/outlet and PV panels temperature. The effect of distance between the panels and rooftop and the solar radiation intensity are two main parameters that were studied. The experimental and numerical results showed that the mean temperature of the PV array with no air gap could be about 12 ± 5 °C higher than one provided with an air gap greater than 200 mm and is about 18 ± 5 °C when the gap increased to 250 mm when the radiation is about 1000 W/m^2. While the efficiency of the PV panel in STC is 18.04%, it drops to 14.17% when the gap is 0 mm and increases to 15.01% and 15.44% for the gaps of 200 mm and 250 mm. CFD simulation shows that surface temperatures of panels are nearly uniform. The CFD simulation follows the experimental results quite closely.

1. Introduction

Due to excessive solar irradiance and high environmental temperatures, PV panels overheating becomes a major problem especially in the regions with hot climate. In real conditions, the temperature of panels increases under direct exposure to solar radiation. Depend on the location of the PV panel installation, the module temperature may reach up to 75 °C which causes up to 25% of power output loss (Abd-Elhady et al., 2018). Therefore, investigating on an effective solution to minimize this power loss is important to researchers and manufacturers.

The various cooling technologies aimed to enhance the efficiency of the solar PV systems and increase the lifecycle of the panels. These techniques are categorized into two categories of active and passive cooling. Passive cooling techniques are mainly used to reduce PV temperature without power consumption, while active cooling techniques use devices such as fans to force air or pump water (Dwivedi et al., 2020). Passive cooling techniques are mainly grouped into air-based, water-based, refrigerant-based, and PCM based techniques. Overall, the passive (active) cooling systems are found to achieve a reduction in PV module temperature in the range of up to 6–20 °C (30 °C) with an improvement in electrical performance up to 15.5% (22%) (Hasanuzzaman et al., 2016).

In a research work, Vaillon et al. (2018) quantified the impact of all possible strategies to minimize thermal losses in the case of the mainstream solar PV panels with crystalline silicon technology. They proposed a general method to analyze every photovoltaic technology on reducing thermal losses. The proposed assessment of pathways for mitigating the thermal losses in the case of crystalline silicon solar photovoltaic panels indicates that there are three main areas to improve the PV panel performances, which are sub-bandgap reflection, radiative cooling, and improvement in conductive/convective cooling capacities. Evidently, efficient ways to minimize thermal losses are to increase

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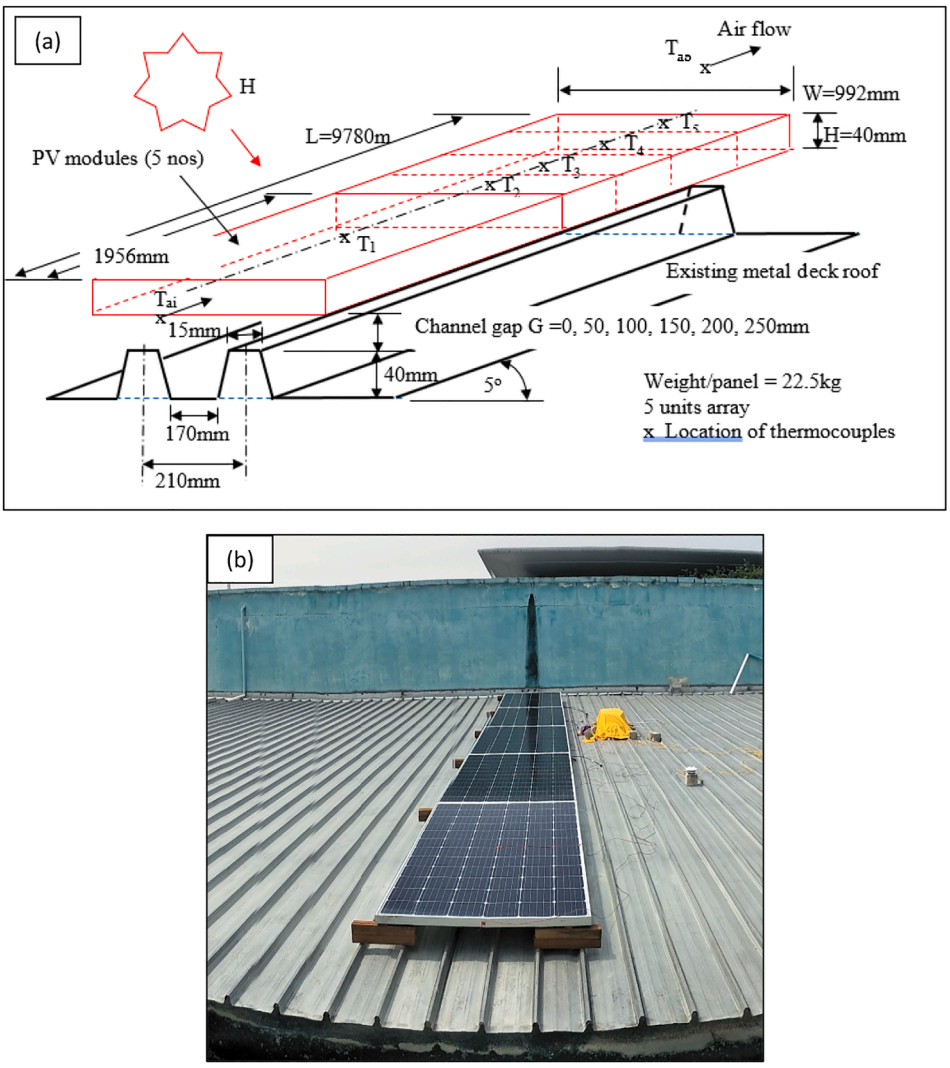


Fig. 1. (a) Schematic, (b) photograph of PV modules set-up over roof with 0 mm gap.

power output in STC and more importantly, to decrease temperature sensitivity.

Due to the strong non-linear relationship between module temperature and convective heat transfer to the air around the PV panel, enhancing the convective cooling could have substantial impacts on module efficiency. Stanislawski et al. (2020) investigated the potential impact of module arrangements on the convective cooling of large PV arrays. Three idealized module arrangements are evaluated in comparison to the traditional, row-organized arrangement. Numerical simulation results indicated that dense arrangements with larger packing parameters more effectively enhance convective cooling than sparse arrangements. Compared to the baseline, the most compact arrangement exhibited an increase in convective heat transfer of 14.8%.

Glick et al. (2020b, 2020c) studied numerically the convective heat transfer coefficient for a utility-scale solar farm. The effects of different directions and configurations of infinite solar PV arrays on the air movement, Reynolds number, and heat transfer around the PV panels were studied. They also carried out combined thermal and particle-image-velocimetry measurements in a scaled wind tunnel experiment. The velocity measurements indicated the complex flow structure within the solar array, governed by wakes directed upwards due to the orientation of the solar panels. It was demonstrated that the convective heat transfer coefficient increases with Reynolds number for both the upper and lower surfaces. They found that increased inflow turbulence intensity increased the Nusselt number on both the front and rear surfaces by as much as 7%. In one other similar work by Glick et al. (2020a), they found that significant changes in the convective heat transfer coefficient are possible, based on wind direction, wind speed, and module inclination. It was found that 30–45% increases in convection are possible through an array-flow informed approach to layout design, leading to a potential overall power increase of ~5% and a decrease of solar panel degradation by +0.3%/year.

Nižetić et al. (2017) evaluated the effect of passive cooling techniques using PCM, air-based, liquid-based, and finally radiative based. They used a 30 kW PV plant as their case study to estimate their positive effect on the levelized cost of energy and life-cycle of the PV system. Their assessment on the existing reports about these techniques shows that technically and economically the air-based solutions with aluminum fins is the most viable one. Cuce et al. (2011) applied aluminum fins behind a polycrystalline PV cell in controlled conditions under solar radiation from 200 to 800 W/m^2, which led to a relative increment of 9% on the performance of the module. Firoozzadeh et al. (2019) used simple and low-cost aluminum fins, which led to a reduction of maximum temperature from 85 °C to 77.6 °C and 2.7% increase in efficiency. Grubišić-Čabo et al. (2018) mounted few aluminum fins on the backside surface of a 50 W PV panel, which in the hottest time of the day causes to 2 W increment in the output of the panel. Abdel-raheim Amr et al. (2019) found that at solar irradiation level of 800 W/m^2, while ambient temperature increase from 25 to 50 °C, power output of a 250 W module decreases from 151.2 to 139.6 W in usual panel against

Table 1
Solar PV panel specifications.

Parameter	Value
Dimension	1956 mm × 992 mm
Maximum power	350 Wp
Panel efficiency	18.04%
Temperature coefficients of Pmax	−0.39%/°C

180.9 to 162.3 W in the backed finned panel. Elbreki et al. (2020) parametrically studied fin height, number fins, fin spacing, wind speed, and solar irradiance on the PV module performance of a combined planar reflector and backplate extended surface. The recorded the temperature difference of over 24 °C, caused to 11.2% increment in the electrical output at 1000 W/m^2.

In a specific design, Sheyda et al. (2014) placed a wind-driven rooftop turbine ventilator beneath the PV module to create airflow under the PV module, as well as power generation and ventilation, by rotation of the turbine. Experimental results show about 15 °C reduction on the module temperature under the radiation of 1000 W/m^2 and wind velocity over 4 m/s. In one other innovative research, Abd-Elhady et al. (2018) drilled a few holes in the PV panel to assist moving hot air upward from the downside of the panel in Egyptian weather conditions. This natural convection simple method shows effective in comparison with the baseline panel. The simulation shows the effect of the number of holes and the optimum diameter (12–19 °C temperature drop).

Using ducts under the PV panel also received attention from researchers. PV modules could be passively cooled by providing an air duct underneath and allowing natural convection air to flow underneath. Brinkworth and Sandberg (2006), Brinkworth (2006) studied natural convection through a duct under the PV panels. They suggested the optimum length to depth (L/D) ratio is about 20 and claimed that the principal parameter for design on the duct is its depth, while the inclination of the roof does not affect the ration that much. Baloch et al. (2015) tested two PV panels, which one is in a normal situation and the other one with a duct channel beneath of that, in the hot climate of Saudi Arabia for the second half of the year. They reported temperature drops from 71 °C to 48 °C and from 45 °C to 36 °C in June and December, respectively, which led to averagely over 35% of power output and a relatively 19% reduction in the levelized cost of energy.

The cheapest and simplest method of PV cooling is using air-cooling with a natural airflow below the PV panels, according to its stack effect. In this method, no specific arrangement between the PV panels is required. However, the effect of heat removal by natural air circulation on the performance of the PV panel is limited. The main reasons are low convective heat transfer, air thermal conductivity, low volumetric heat capacity, and low density.

Pitched metal deck or clay tile roofs are very popular in many countries. The installation of the PV panels on these roofs typically requires aluminum frames as a mounting structure. One important question is what would be the most optimum gap between the PV panel and roof for the PV panel installation to maximize the natural heat convection. The objectives of this experimental and numerical investigation are to determine the surface temperature in an array of PV modules mounted over an existing inclined metal deck roof with various airflow channel gaps underneath.

2. Experimental setup and procedure

Five Jinko monocrystalline PV panels (Model: Eagle PERC 72 M 350 W) were laid out in series over an existing 5°-inclined metal deck roof with a trapezoidal profile as shown in Fig. 1a. Specifications of panels are provided in Table 1. The total length of the 5 panels array is 9.780 m. The panels were laid out on top of the roof purlins and supported with timber struts corresponding to the required channel gap (G). The channel gap was assumed to be between the bottom lip of each PV panel

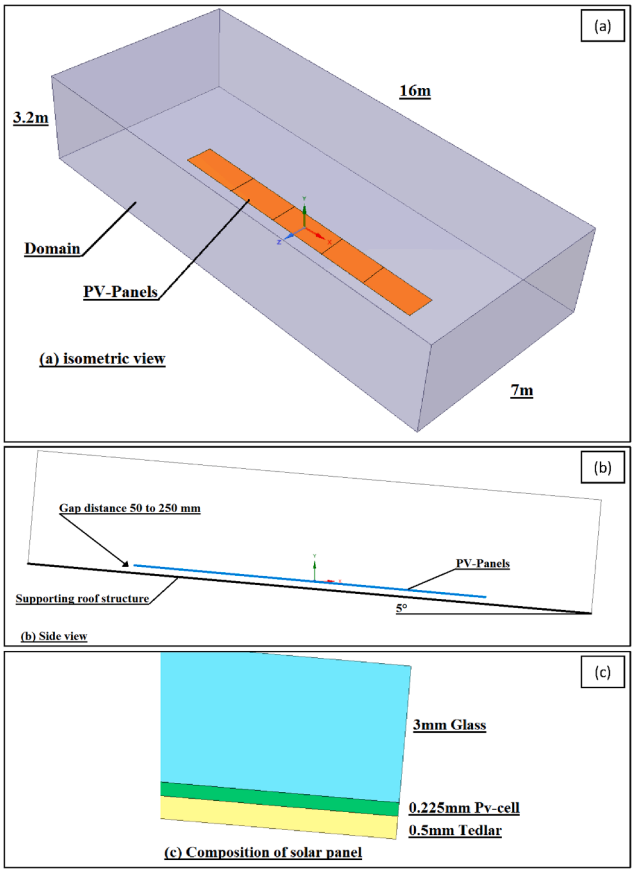


Fig. 2. (a) Simulation model of PV panels and domain size, (b) side view, (c) solar PV components.

and the upper surface of the purlins as shown in Fig. 1a. The inlet edge of the first panel was set back about 0.5 m from the edge of the roof. A photograph of the experimental layout and the support assembly with G = 50 mm is shown in Fig. 1b.

Copper-constantan thermocouples (Type T) with an accuracy of ± 0.5 °C were employed to measure temperatures. Five thermocouples were attached on the top surface of each panel at their mid-points (T_1, T_2, T_3, T_4, and T_5). Also, two thermocouples ($T_{a,i,}$ and $T_{a,o}$) were located at the inlet and exit sections of the array to measure inlet and outlet induced air temperatures. Solar radiation (H) was measured with a Meteon solarimeter, which is placed in the rooftop nearby the panels.

The Left and right sides of the panels are not covered. However, the data collection carried out in days that there were not any considerable wind blowing around the panels.

3. Numerical model

In this section, the structure of a PV module, and the governing equations used in the simulation process along with the thermophysical and optical properties of all sections of the PV panel are expressed.

Table 2
Thermophysical properties of PV panel (Hamrouni et al., 2008).

Layer	Thickness (m)	Thermal conductivity (W/mK)	Density (kg/m^3)	Specific heat capacity (J/kgC)
Glass	0.003	1.8	3000	500
Pv Cells	0.000225	148	2330	677
Tedlar	0.0001	0.2	1200	1250

Table 3
Optical properties of PV panel components for normal incidence light (Bhattarai et al., 2012).

Layer	Parameter	Value
Glass	Emissivity	0.88
	Absorption coefficient	26 for < 4.25 μm
PV cells	Refractive index	1.526
	Emissivity	0.85 for 4.25 μm
Tedlar	Refractive index	1.45

where S is the absorbed solar flux, and V, A, and T are the volume, area, and average temperature of the PV module.

á¶¯$_{ref}$, T_{ref} and β_{ref}, are the electrical efficiency, temperature of the PV panel, and temperature coefficient in reference mode. To in Eq. (5) is the temperature at which the PV panel's electrical efficiency drops to zero.

Momentum Equation:

In the fluid section, the velocity vector u and Q = 0 should be found by solving the continuity and momentum equations for the incompressible wind flow.

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)=-\frac{\partial P}{\partial x}+\frac{\partial}{\partial x}\left[\mu\left(2\frac{\partial u}{\partial x}-\frac{2}{3}\nabla.\vec{V}\right)\right]+\frac{\partial}{\partial y}\left[\mu\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)\right]+\frac{\partial}{\partial z}\left[\mu\left(\frac{\partial u}{\partial z}+\frac{\partial w}{\partial x}\right)\right]+B_x$$
$$\rho\left(\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\right)=-\frac{\partial P}{\partial y}+\frac{\partial}{\partial y}\left[\mu\left(2\frac{\partial v}{\partial y}-\frac{2}{3}\nabla.\vec{V}\right)\right]+\frac{\partial}{\partial x}\left[\mu\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)\right]+\frac{\partial}{\partial z}\left[\mu\left(\frac{\partial v}{\partial z}+\frac{\partial w}{\partial y}\right)\right]+B_y \quad (7)$$
$$\rho\left(\frac{\partial w}{\partial t}+u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right)=-\frac{\partial P}{\partial z}+\frac{\partial}{\partial z}\left[\mu\left(2\frac{\partial w}{\partial z}-\frac{2}{3}\nabla.\vec{V}\right)\right]+\frac{\partial}{\partial x}\left[\mu\left(\frac{\partial w}{\partial x}+\frac{\partial u}{\partial z}\right)\right]+\frac{\partial}{\partial y}\left[\mu\left(\frac{\partial v}{\partial z}+\frac{\partial w}{\partial y}\right)\right]+B_z$$

3.1. Physical model

Fig. 2a and 2b show the simulation model of the experimental setup. A PV module consists of glass in the outer layer, the monocrystalline silicon cells (PV cell), and a Tedlar polyvinyl fluoride layer (Tedlar), as shown in Fig. 2c. All these layers along with the roof structure and air gap were considered.

The thermophysical properties and optical properties of PV panel components are listed in Table 2 and Table 3.

3.2. Mathematical formulation

The steady-state energy and momentum equations are solved to find the temperature in the entire domain Area. In Cartesian coordination and taking into account of the simplifying assumptions explained in the next section, the governing equations for the heat transfer problem are as follows:

Energy equation:

In the following equations, Q is the volumetric heat source and u is the velocity vector. Obviously, the velocity vector is zero in the solid PV panel area, so Eq. (1) cuts to Poisson's equation with the two basic terms of Q: (1) the absorbed solar radiation modeled as a positive volumetric source; (2) the electrical output power modeled as a negative term Eq. (2).

$$\rho c_p u.\nabla T+\nabla.q=Q, \quad (1)$$

$$q=-K\nabla T. \quad (2)$$

Conservation of Mass:

$$\frac{\partial \rho}{\partial t}+\nabla.\rho\vec{V}=0 \quad (3)$$

Thermophysical properties are assumed to be constant.

$$Q=\frac{-P_{out}+S.A}{V} \quad (4)$$

$$P_{out}=\text{á¶¯}_{ref}\left(1-\beta_{ref}\left(T-T_{ref}\right)\right)S.A \quad (5)$$

$$\beta_{ref}=\frac{1}{T_o-T_{ref}} \quad (6)$$

Conservation of Energy:

$$\rho\left(\frac{\partial E}{\partial t}+u\frac{\partial E}{\partial x}+v\frac{\partial E}{\partial y}+w\frac{\partial E}{\partial z}\right)=\nabla.(k\nabla T)-\nabla.p\vec{V}+\dot{Q}_v+\dot{Q}_g \quad (8)$$

Equation of State:

$$\rho=\rho(P.T) \quad (9)$$

Therefore, the system of Eqs. (1)–(9) should be solved simultaneously.

3.3. Boundary conditions and assumptions

Metal roofing is made from a grid of steel available in a range of coating either Galvanized steel. Its special anti-capillary features alongside lap make it suited for severe weather conditions. In reality, an aged metal roof may have some corrosion and this causes turbulent air movement. However, the setup assembled in a location that there was no corrosion on the roof. Therefore, laminar airflow was considered for the simulation calculations. Both convention and radiation are applied to the external surface of the glass cover. The boundary condition considered for all walls exposed to the airflow is no-slip condition. The temperature of external radiation is considered to be equal to the sky temperature, which is determined by the following equation (Swinbank, 1963):

$$T_{sky}=0.0552\times T_{amb} \quad (10)$$

The following input parameters are assumed in this modelling:

- $\alpha_{pv}=\varepsilon_{pv}=0.53$
- $u_{wind}=0$
- $T_{amb}=30\ ^{\circ}C$
- H_{rad} = solar radiation intensity according to the experimental data
- T_{ai} = air inlet temperature according to the experimental data
- Solar radiation is limited to the direct radiation.
- The solar radiation is perpendicular to the panel surface.
- The thermal properties of PV panel components are independent of the temperature.

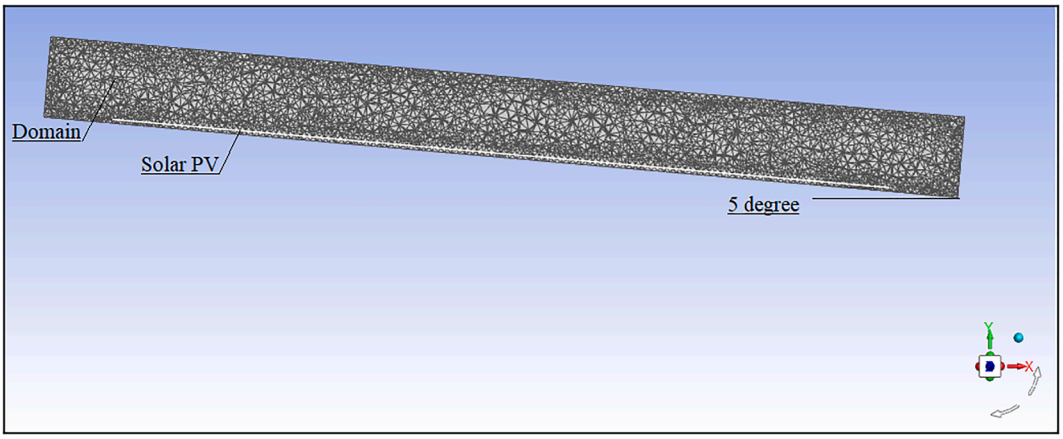


Fig. 3. The mesh used in the present simulation.

Table 4
The grid study results.

Number of elements	Average temperature on PV panel (C)	Maximum temperature on PV panel (C)
160,991	50.1	52.2
195,910	50.8	52.7
230,510	51.9	53.2
259,229	52.7	53.1
330,519	52.8	53.1

- The boundary condition for sidewalls of the PV panels and the bottom of the supporting roof structure is considered as adiabatic (shown in Fig. 1b).
- The thermal heat gain of air at the top of the PV panel is neglected.
- The attachment between layers of the PV panel is the perfect thermal contact.

Zondag et al. (2002) experimentally proved that in a nearly clear sky using hourly data can provide sufficiently accurate results for the calculation of the performance of PV panels. This can help to reduce the numerical simulation processing time. Therefore, due to the convergence of the results in each time step, the proposed steady-state model in this study can be considered as a quasi-steady model too.

3.4. Numerical solution

ANSYS Fluent 19.1 is used as a CFD package for all the simulations performed in this study. A CFD model was developed to solve the governing equations. In this laminar model, the pressure-based solver and coupling of pressure and SIMPLE scheme were used for velocity (Patankar, 1980). For the convection, diffusion, and radiation terms, the "second-order upwind" was selected and the pressure PERESTO scheme was selected. The ray-tracing algorithm was used for the sun's rays modeling that enters the computational domain (Peyret, 1996). The ray-tracing algorithm is coupled with the energy equation via a heat source term. The normalized convergence criterion for the continuity, momentum, and DO equations were set 10^{-6} and 10^{-8} was used for the energy equation.

The computational domain is adopted with 3D structured grids and refined grids near the walls, which is presented in Fig. 3. Appreciations to using the lower cell numbers, and a pre-specified pattern to sort the nodes, structured grids require less computational memory and CFD computation time compared to the alternative unstructured grids.

Table 4 shows the evaluation of the dependence of results on the mesh. The grid study of the results is checked by growing the mesh numbers. The results show that 2.5×10^5 cells are appropriate for obtaining satisfactory results.

4. Results and discussion

Results are presented in two sections under experimental and numerical. Experimental tests were carried out in the actual field environment. Due to a slight difference between the boundary and initial conditions of these two methods, the result presented separately. However, the extracted results provided similar meanings.

4.1. Experimental results

Six sets of temperature and radiation data collection were obtained for five gaps (G) between panels and rooftop (0 mm, 50 mm, 100 mm, 150 mm, 200 mm). Each Run was recorded for four hours, from 10:00 until 14:00. Experimental data were recorded every 5 min. Typical data were compared with each other and are presented in Table 5.

Solar radiation, ambient temperature, and wind conditions fluctuated throughout the day from sunrise to sunset. The angle of solar incidence also changes daily. Temperature increases as radiation increase as expected. Wind and thermal inertia play very important roles

Table 5
Data for the selected runs.

#Run	G (mm)	Date	Weather condition	10:00	11:00	12:00	13:00	14:00
1	0	16/09/2019	Ambient Temp.	39.2	36.3	46.8	41.4	41.4
			Day mode	Partial sunny	Partial sunny	sunny	sunny	sunny
2	50	21/09/2019	Ambient Temp.	33.5	39.7	39.5	42.0	41.9
			Day mode	Sunny	Sunny	Sunny	Sunny	Sunny
3	100	2/10/2019	Ambient Temp.	30.6	35.9	41.3	40.4	38.7
			Day mode	Cloudy	Sunny	Sunny	Sunny	Cloudy
4	150	10/10/2019	Ambient Temp.	29.7	34.6	37.5	38.5	41.9
			Day mode	Sunny	Sunny	Sunny	Sunny	Sunny
5	200	19/10/2019	Ambient Temp.	30.5	32.3	35.3	36.2	36.9
			Day mode	Cloudy	Sunny	Sunny	Sunny	Sunny

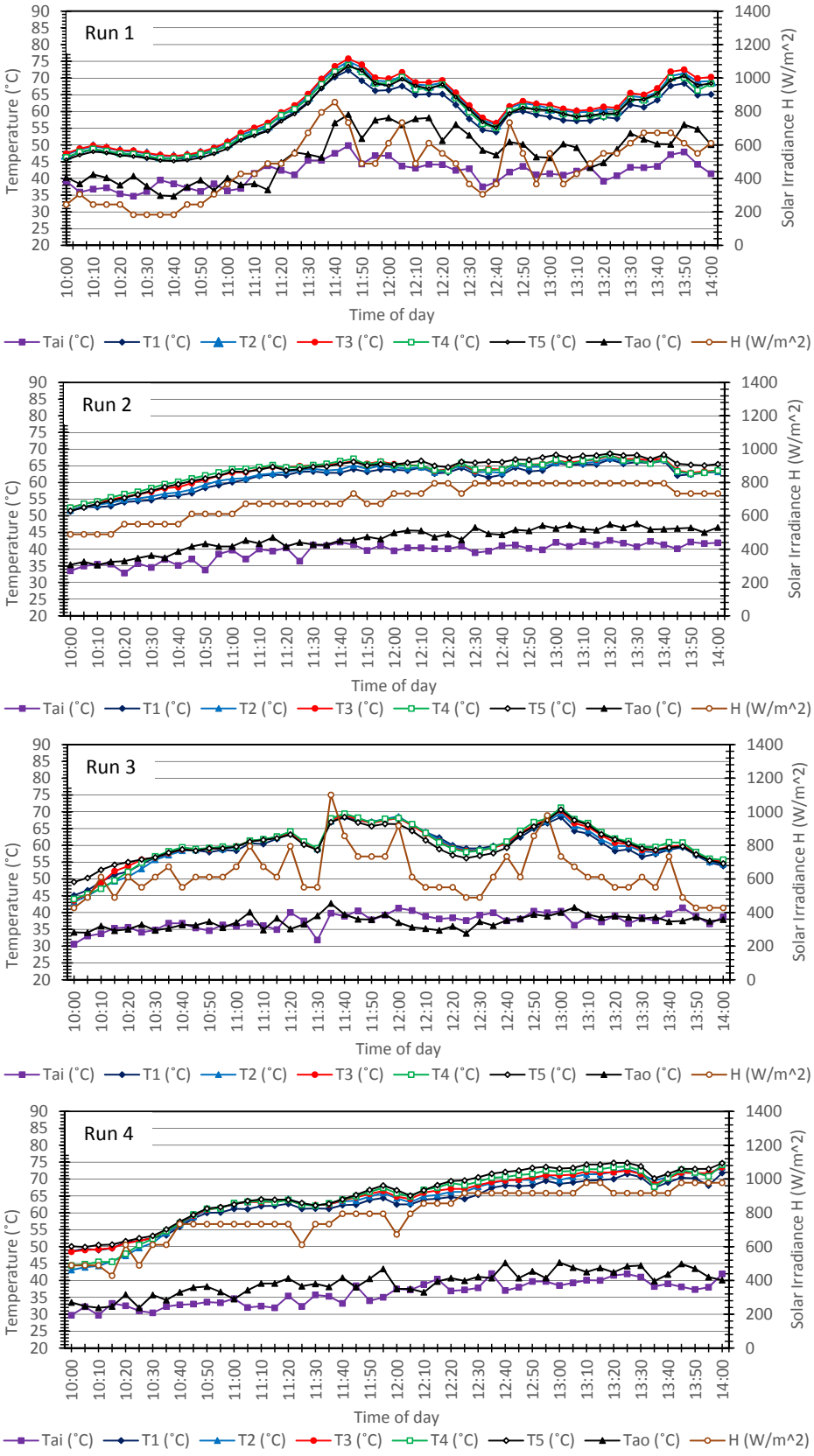


Fig. 4. Daily transient PV panel temperature, inlet/outlet air temperature and radiation vs time for Runs #1 to #5.

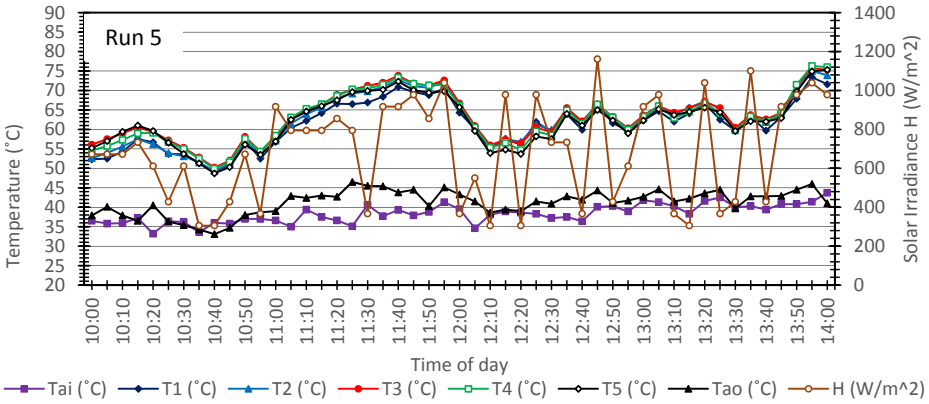


Fig. 4. (*continued*).

in determining surface temperatures. It is impossible to pinpoint the individual roles played by the various factors as daily conditions are not always the same. It was observed that at different times of the day, even with the similar value of radiation, the inlet air temperature (T_{ai}) could vary from about 41.9–49.8 °C due to the heating up of the air by the roof deck before entering the PV array. If the first panel was laid out at the edge of the roof, the inlet air temperature would correspond to the ambient air temperature. This fact causes a difference in PV panel temperature. For example, the temperature of the middle panel (T_3) varied from 61.5 to 75.8 °C.

All temperatures increase with radiation as expected. Inlet air temperature also increases due to the warming up of the air by the roof before it enters the PV panel. In Fig. 4 for Run #1 with no gap (G = 0) the first panel (T_1) exhibited the lowest temperature while the middle panel (T_3) showed the highest temperature. Temperatures T_2, T_4, and T_5 are quite uniform. At a radiation level of 800 W/m^2, the temperature difference between the first and third panel was about 4–5 °C and the induced air temperature difference between inlet and outlet was about 10 °C. For Run #2 with an air gap of G = 50 mm, graphs show that at H = 800 W/m^2 the temperature difference between the first and second panel were lowest and that panels 3, 4, and 5 were about equal. The induced air temperature difference was about 4–5 °C. Fig. 4 shows that generally, all the 5 panels were at a quite close temperature at all radiation levels. The maximum temperature difference between inlet and outlet air is in run #5 when the gap is 200 mm and solar radiation reaches about 1000 W/m^2. This fact shows that while the gap increases in higher solar radiation the heat picks up from the panels increase, and it leads to less increment in the panel surface temperature.

Fig. 5 shows the typical temperature distribution obtained at the various values of H for G = 0, 50, 100, 150, and 200 mm, respectively. Linear regression lines of temperatures of all panels vs radiation (H) are obtained and then, PV panels temperature distribution is cross-plotted against solar radiation. By increasing the radiation intensity, the temperature difference between inlet and outlet points also increases in almost all of the runs.

Fig. 6 presents PV panels temperature and air inlet/outlet fluctuations for different gaps in specific radiations. The effect of gaps along the PV array in specific radiation indicates the influence of underneath natural convection on the temperature of the panels. The variation in temperatures obtained due to wind, ambient temperature, and inertial effects could result in temperature variations of 2 °C or greater. Bearing in mind that due to prevailing weather conditions results are “repeatable” to as high as 5 °C, it is generally observed that:

1) All temperatures increase with radiation.
2) The highest mean temperatures are achieved with no air gap (G = 0).
3) The lower temperature was achieved with G from 50 mm onwards.
4) At 800 W/m^2, the mean array temperature with G = 0 mm was about 73 °C compared to 61 °C with G = 200 mm, which shows 12 °C temperature reduction.

4.2. Simulation results

A three-dimensional simulation model was proposed as shown in the isometric view of Fig. 2. The sides of the domain were assumed to be symmetrical. The air gap between the supporting roof structure at the bottom of the domain and the solar panels on the top of the roof varied from 0 to 300 mm. Using a finer mesh with high resolution did not show any change in the results obtained.

4.2.1. Validation

To validate the accuracy of the numerical results, the panel temperatures were compared with the experimental temperature measurement of PV panels. As indicated in Fig. 7, numerical and experimental data follow a similar pattern. The average difference between the two results is about 5 °C. The main reason for this difference is due to the simplifying assumptions that were used in the simulation, uncertainties of experimental data, fixed solar radiation in simulation, fixed inlet air temperature, and of course not considering the air movement from the sides of the PV array. Hence, the proposed three-dimensional model is accurate enough to simulate PV panel temperature fluctuations.

4.2.2. Effect of the gap distance

Fig. 8 shows the CFD temperature distribution obtained for different gap sizes from 0 to 300 mm with H_{rad} = 1000 W/m^2 and T_{amb} = 30 °C. Uniform temperature distribution was observed along with the five panels. The surface temperature decreases with increasing air gaps. The temperature difference with the gap of 200 mm is about 14 °C. There are significant temperature differences, about 18 °C with gaps from 250 mm and 300 mm. the temperature difference for the gaps of 250 mm and 300 mm is very minor. A possible explanation is due to the varying effects of convection and radiation heat transfer between the supporting roof and the solar panels.

4.2.3. Effect of radiation

Simulations are carried out at specific radiation levels (H_{rad}) and fixed inlet air temperatures (T_{ai}) to determine the temperatures at the midpoints of each of the five panels (T1, T2, T3, T4, T5) for each air gap (G). The results between simulated and actual experimental data show that the panels displayed quite uniform temperatures.

Fig. 9 shows the simulations for five gaps of 50, 100, 150, 200, and 250 mm at solar radiation levels of 400, 600, 800, and 1000 W/m^2. The graphs show that there is no significant difference in temperatures for gaps between 50 and 200 mm at H = 400 W/m^2 (about 4 °C). However, the temperature difference between the gaps of 50 and 250 mm

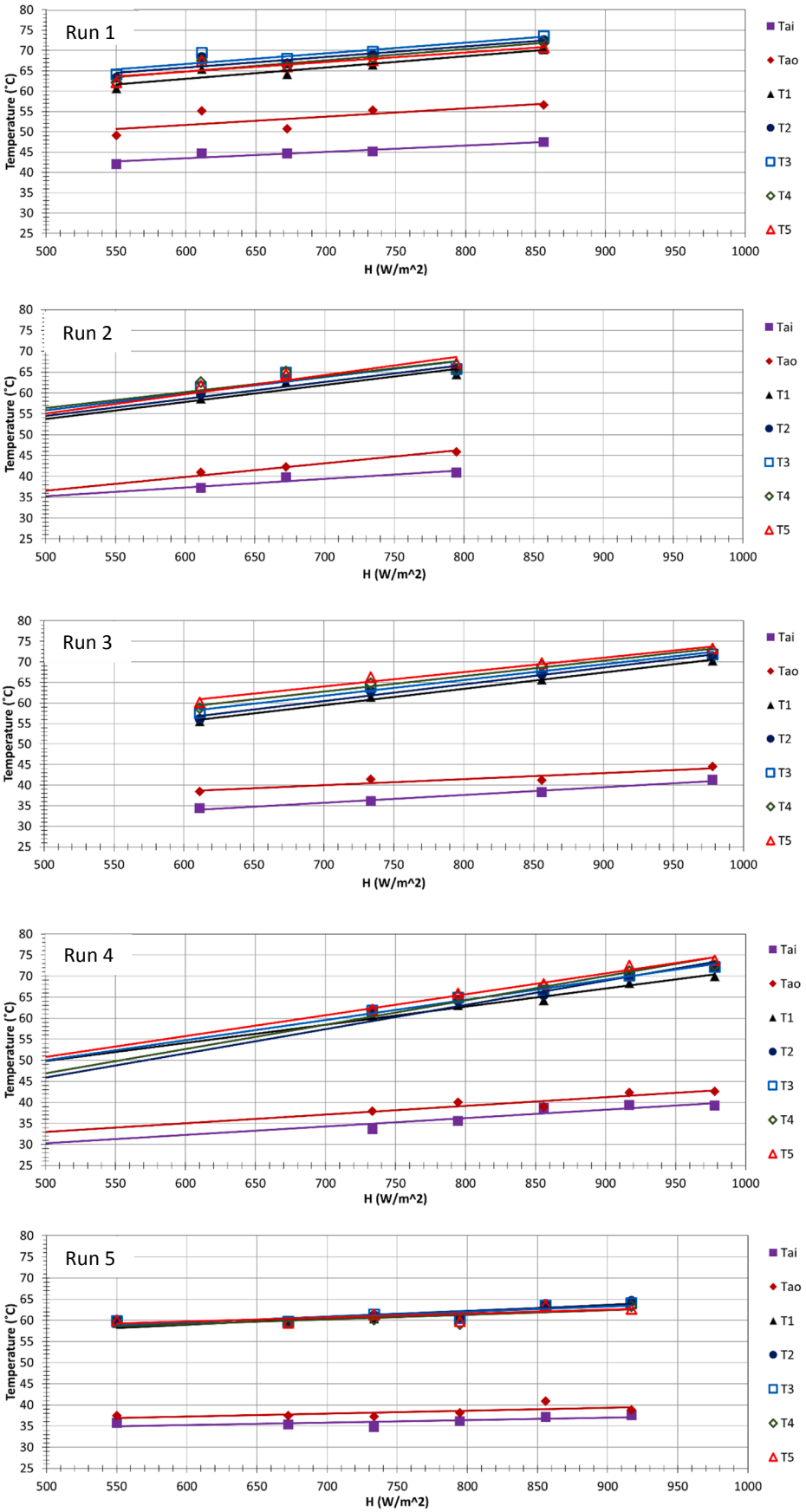


Fig. 5. PV panels temperature vs radiation for Runs #1 to #5.

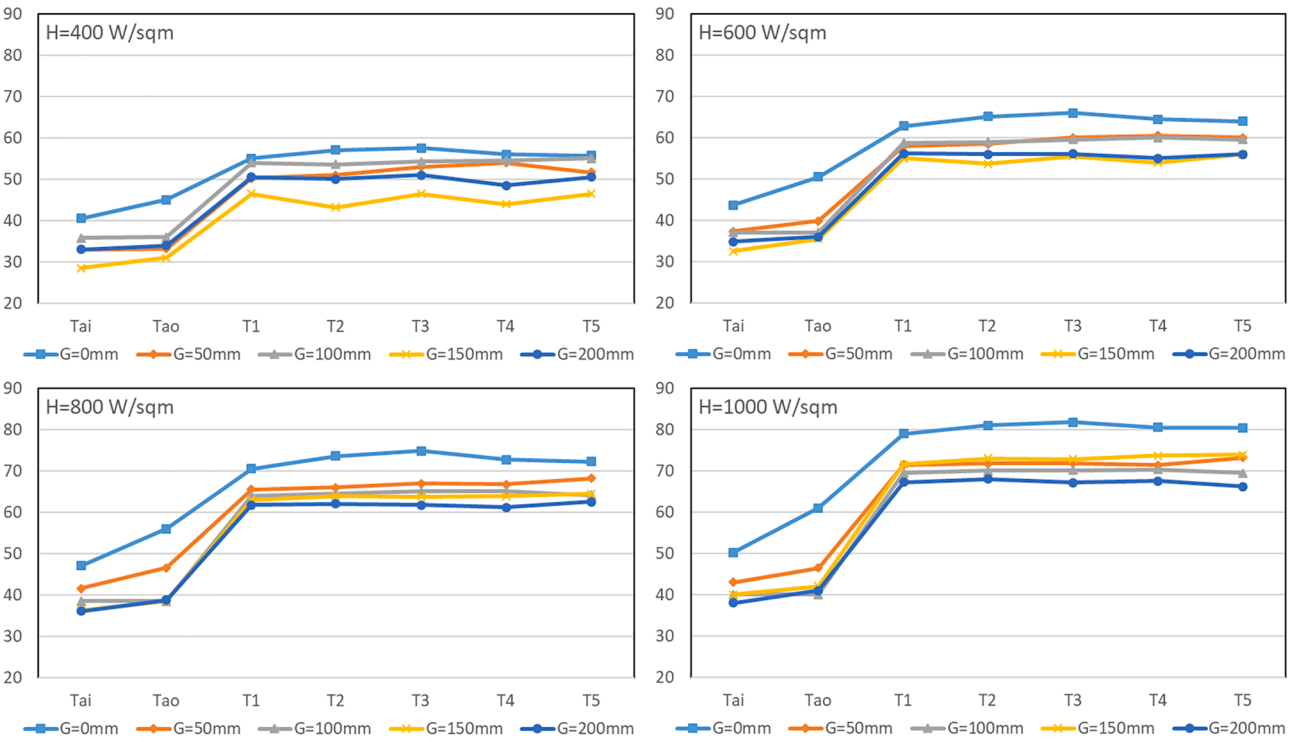


Fig. 6. Typical PV panel temperature fluctuations, air inlet/outlet for different gaps in specific radiation.

amounted to about 10 °C. There is a negligible difference in temperatures difference for 50 and 150 mm gaps for radiations of 600 and 800 W/m^2. At 1000 W/m^2, the temperature differences decreased with increasing gaps. It should be highlighted that these results have a direct proportion to the imposed inlet air temperature T_{ai} and also fixed ambient air temperature (T_a assumed = 30 °C).

Simulation results show a uniform temperature distribution along with the panels array, which is close to the experimental results. The surface temperature decreases with increasing air gaps.

Fig. 10 graphically presents the temperature distribution along with the panels array in the 3D view, side view of the array, and the tip of the last panel. Unsurprisingly, a non-uniform temperature distribution is observed in the PV module. As it is indicated, the hottest point in the array is the last panel, which is expectable. The entry air underneath of the panels warms up by passing under the array. The temperature distribution of the middle plan of domain and PV panels illustrated in Fig. 10(down). The higher area of the domain equal to the ambient temperature (30 °C) and the higher temperatures range shows at the top of the PV panels due to the low velocity. Due to the nonuniform heat transfer coefficient distribution the complex velocity field that occurs around the PV panels as shown in Fig. 11. The high-velocity wake shaped behind the PV panels is responsible for the convection heat transfer coefficient and the ensuing higher temperatures in this area. These figures are at the radiation of 1000 W/m^2 and the gap of the 250 mm and inlet wind velocity of 0 m/s.

By using back Eq. (5) and findings of temperature reduction on panels with different Gaps, the performance increment of the panels could be estimated. The temperature reduction in gaps 200 mm and 250–300 mm causes to 12 °C (experimental) and 18 °C (simulation), respectively. The average temperature of panels in 1000 W/m^2, when the gap is 0 mm is above 80 °C. While the efficiency in STC is 18.04%, it drops to 14.17% when the gap is 0 mm and increases to 15.01% and 15.44% for the gaps of 200 mm and 250 mm.

5. Conclusion

Installation of the PV panels on a pitched metal deck or clay tile roofs typically requires aluminum frames as a mounting structure. The optimum gap between the panels and the roof maximizes the natural heat convection. The effect of the natural convection on PV panel temperature reduction by free air movement beneath 5 PV panels was experimentally and numerically studied. Temperature measurements of air inlet/outlet and PV panels are presented. The effect of distance between the panels and rooftop and the solar radiation intensity are two main parameters that were studied. The experimental studies were carried out from 0 mm to 200 mm. However, the numerical simulation continued the study up to 300 mm.

The experimental results showed that the mean temperature of the PV array with no air gap could be about 12 ± 5 °C higher than one provided with an air gap greater than 200 mm and is about 18 ± 5 °C when the gap increased to 250–300 mm in the numerical study. All the 5 panels were at a quite close temperature at all radiation levels. By increasing the radiation intensity, the temperature difference between inlet and outlet points also increases in almost all of the runs. At 1000 W/m^2, the mean array temperature with G = 0 mm was about 80 °C compared to 68 °C with G = 200 mm, which shows 12 °C temperature reduction.

CFD simulation shows that uniform temperature distribution was observed along with the five panels at all radiation levels. Surface temperatures decrease with increasing air gaps and radiation intensity. Significant temperature differences are observed with gaps greater than about 200 up to 300 mm due to possible interacting effects of convection and radiation heat transfer between roof and PV panels. The CFD simulation follows the experimental results quite closely. While the efficiency of the PV panel in STC is 18.04%, it drops to 14.17% when the gap is 0 mm and increases to 15.01% and 15.44% for the gaps of 200 mm and 250 mm.

For future works, it could be recommended to consider different types of rooftop materials and the inclination of the roof.

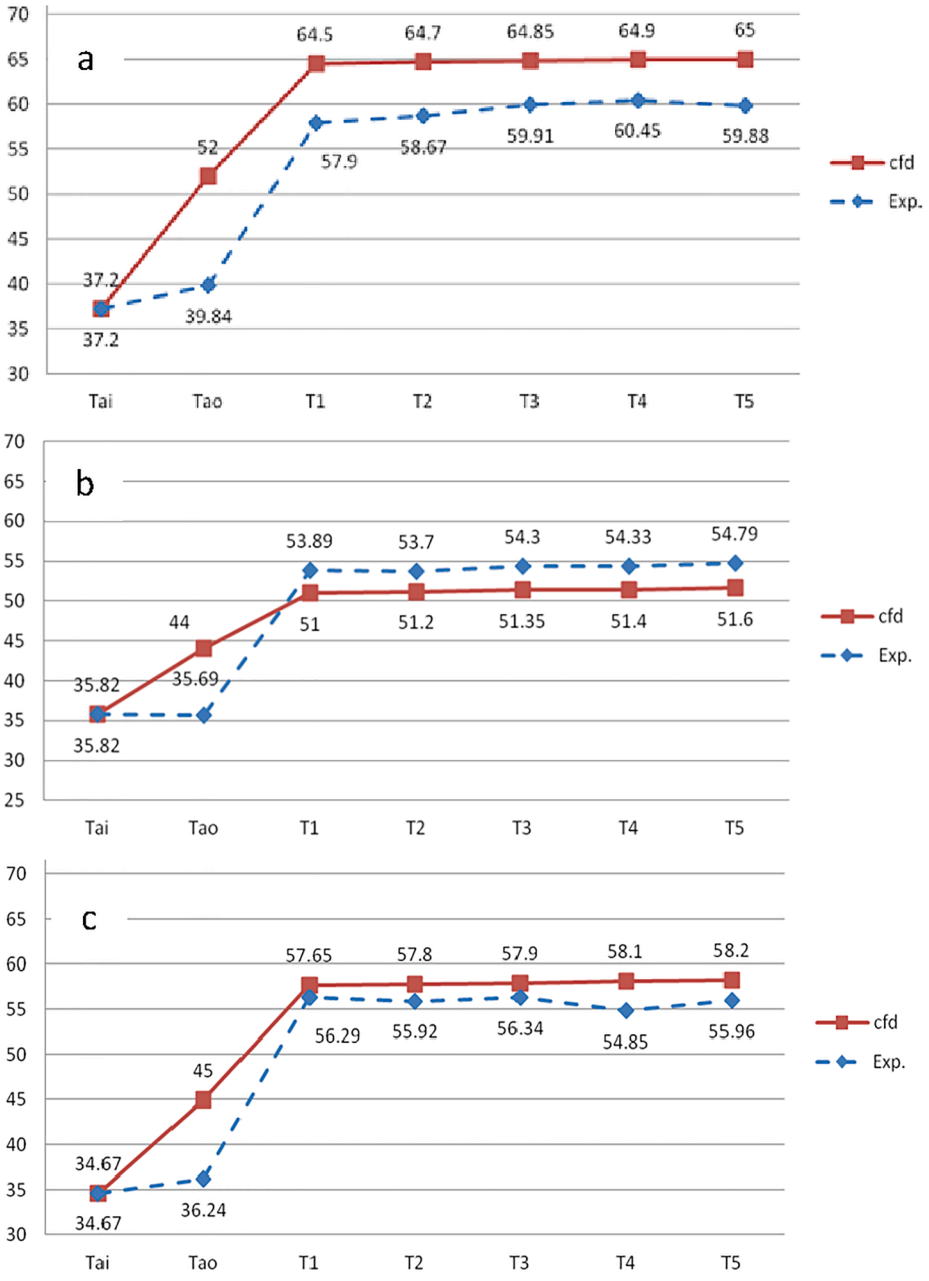


Fig. 7. CFD vs experimental results for (a) Run #2 (G = 50 mm, H = 600 W/m^2), (b) Run #3 (G = 100 mm, H = 400 W/m^2), (c) Run #5 (G = 200 mm, H = 600 W/m^2).

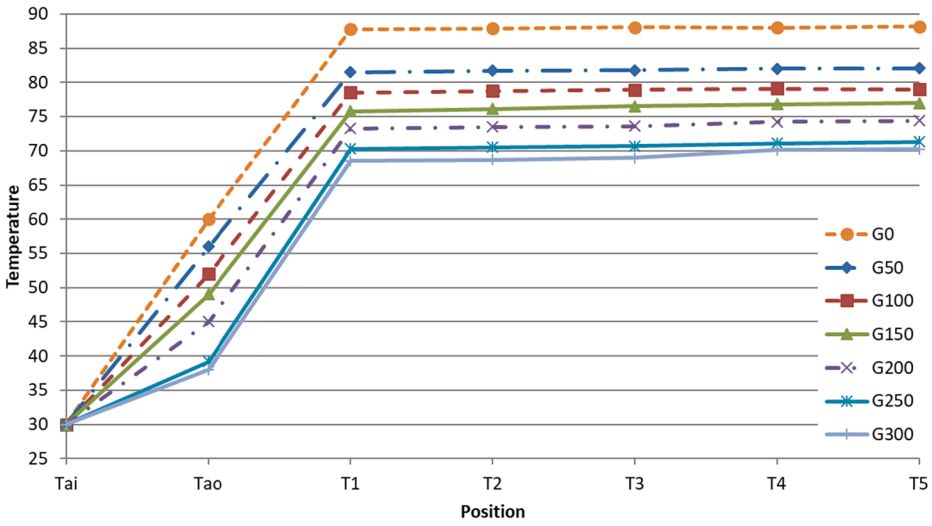


Fig. 8. CFD simulation of temperature distribution with H = 1000 W/m^2 and Tamb = 30 °C.

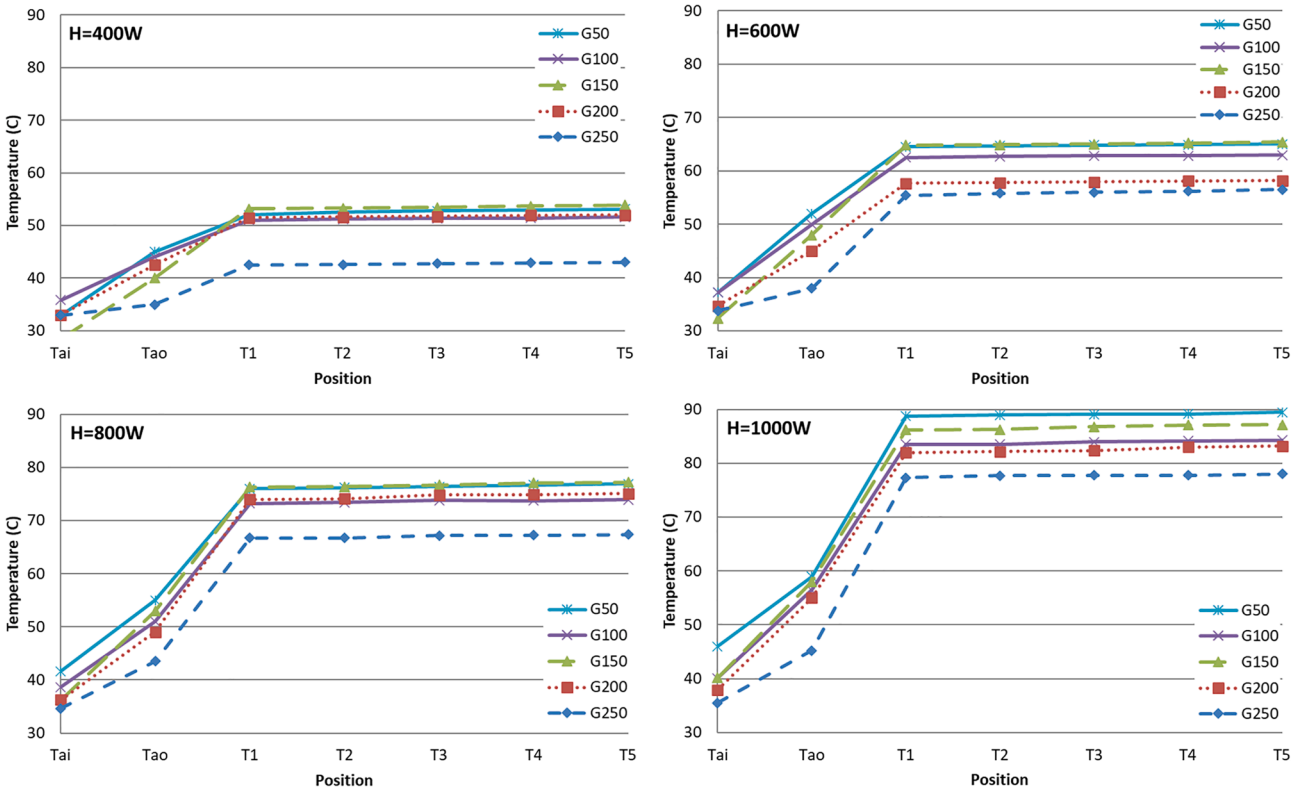


Fig. 9. Fluctuation of the mean temperature of PV panels for different gaps in four solar radiation.

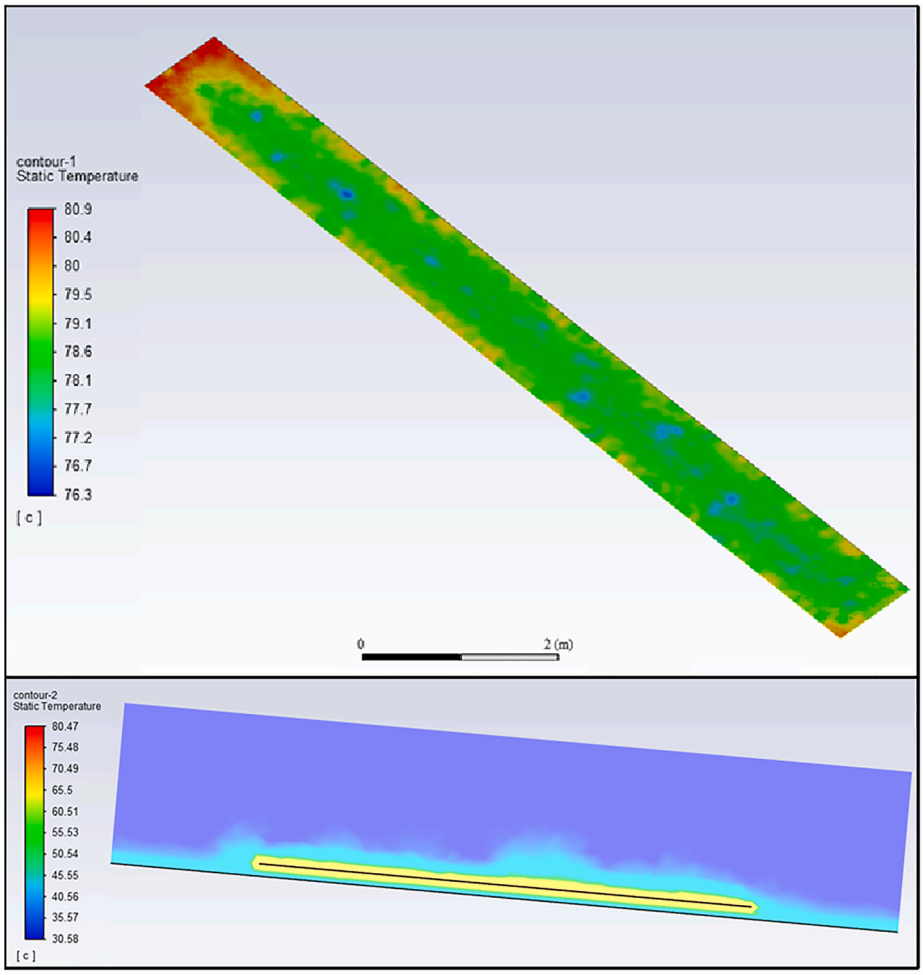


Fig. 10. Temperature distribution along PV array while H = 1000 W/m^2 and T_{amb} = 30 °C.

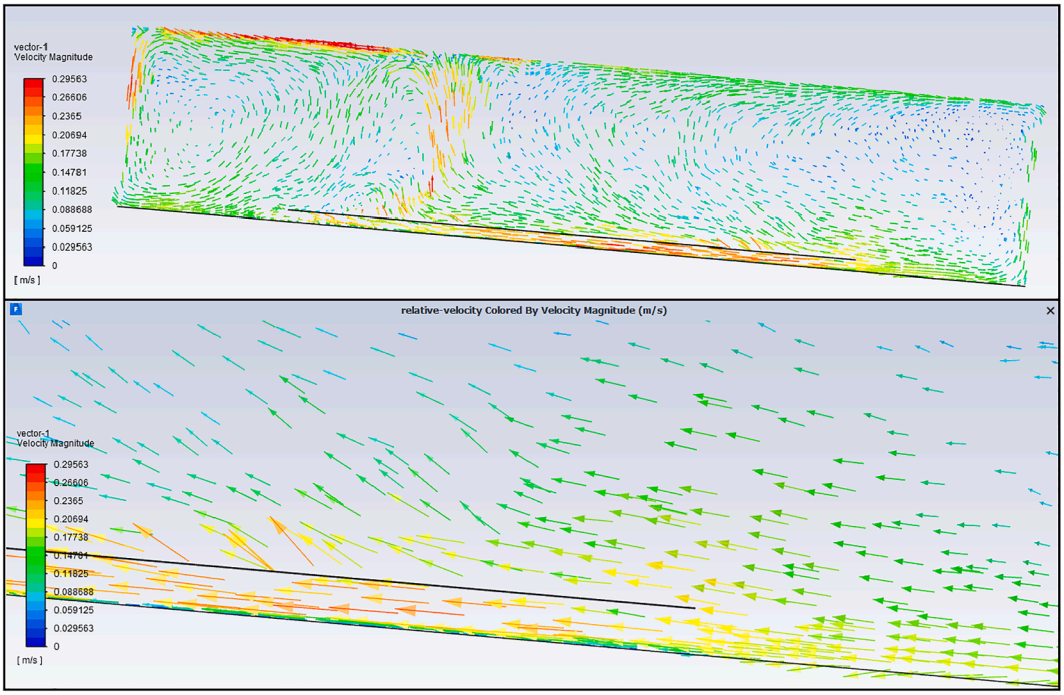


Fig. 11. Velocity Magnitude on middle plan of domain and PV panels.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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